AGA KHAN UNIVERSITY EXAMINATION BOARD
SECONDARY SCHOOL CERTIFICATE CLASS IX EXAMINATION

Chemistry Paper I (Theory & ATP) — Annual 2021

Total Time: 1 hour 40 minutes | Total Marks: 50 (40 Theory & 10 Alternate to Practical)

INSTRUCTIONS: Answer all fifty multiple choice questions. Choose ONE best answer for each question.

Q.1 If hydrogen is replaced with deuterium in each of the given compounds, then the formula of which of these is written INCORRECTLY?

A. DF

B. ND3

C. MgOD

D. C2D5OD

Q.2 How many atoms of chlorine are there in 250 g of chlorine gas? (Note: 35.5_17 Cl)

A. 2.12 × 10^24

B. 4.24 × 10^24

C. 4.43 × 10^24

D. 8.85 × 10^24

Q.3 Which of the following chemical equations is balanced?

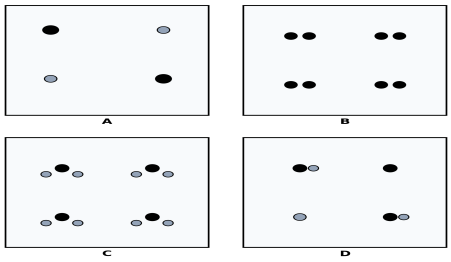
A. CH4(g) + O2(g) -> CO2(g) + H2O(g)

B. NaCl(aq) + BeF2(aq) -> NaF(aq) + BeCl2(s)

C. AgNO3(aq) + LiOH(aq) -> AgOH(s) + LiNO3(aq)

D. Al2(SO4)3(aq) + CaCl2(aq) -> AlCl3(aq) + CaSO4(s)

Q.4 Which of the following diagrams represents a pure compound?



A. Diagram A

B. Diagram B

C. Diagram C

D. Diagram D

Q.5 AlCl3(aq) + NaOH(aq) -> Al(OH)3(s) + NaCl(aq). To balance the given equation, the coefficient '3' should be placed before

A. AlCl3(aq) and NaCl(aq)

B. NaOH(aq) and NaCl(aq)

C. AlCl3(aq) and Al(OH)3(s)

D. NaOH(aq) and Al(OH)3(s)

Q.6 A compound X has an empirical formula C4H9 and its molecular mass is 114 g/mol. Which represents the molecular formula of compound X? (C=12, H=1)

A. C6H11

B. C7H12

C. C8H18

D. C12H27

Q.7 Molecular formula of hexane is C7H16 and that of calcium carbonate is CaCO3. One mole of these substances contains the same:

A. type of atoms.

B. type of bonding.

C. number of atoms.

D. number of grams.

Q.8 The species that does NOT carry an electrical charge is a/ an

A. anion.

B. cation.

C. proton.

D. neutron.

Q.9 Which of the following statements is TRUE for 40_18 Ar?

A. It contains an equal number of protons and neutrons.

B. It contains more number of neutrons than electrons.

C. It contains more number of electrons than protons.

D. It contains more number of protons than neutrons.

Q.10 In Rutherford's experiment which led to the discovery of atomic nucleus, the alpha particles were bombarded on a thin gold foil. Each alpha particle consists of:

A. one proton and one neutron.

B. one proton and two neutrons.

C. two protons and one neutron.

D. two protons and two neutrons.

Q.11 A metal compound has the formula XCl2 and a relative formula mass of 95.211. The relative atomic mass of X is (Cl=35.5)

A. 24.211

B. 59.711

C. 61.211

D. 78.211

Q.12 Which of the following statements is TRUE about Bohr's atomic model?

A. It shows that an atom produces continuous spectrum.

B. It proves that the energy of each orbit in an atom is quantised.

C. It proves that the plum-pudding model of an atom was correct.

D. It shows that electrons emit radiations when revolving around the nucleus.

Q.13 An atom of hydrogen contains one proton and zero neutrons. Suppose a neutron is added to the nucleus of hydrogen atom. This addition will cause:

A. an increase in positive charge on the nucleus.

B. the hydrogen atom to convert into one of its isotopes.

C. an increase in the atomic number of hydrogen atom.

D. conversion into an atom of a different element.

Q.14 In contrast to group II elements, group I elements are less

A. hard.

B. reactive.

C. electropositive.

D. metallic in character.

Q.15 Which periodic trend is observed in moving down group 17 of the modern periodic table?

A. Decreasing reactivity

B. Decreasing boiling point

C. Increasing electronegativity

D. Increasing ionisation energy

Q.16 An element has atomic number 14. In which period of the periodic table will it be placed?

A. 1

B. 2

C. 3

D. 4

Q.17 The horizontal rows in the modern periodic table are known as

A. shells.

B. blocks.

C. groups.

D. periods.

Q.18 According to the modern periodic law, the elements in the periodic table are arranged as per their

A. proton number.

B. neutron number.

C. nucleon number.

D. valence shell electrons.

Q.19 The table shows difference between X and Y families: X (Unreactive, Monoatomic), Y (Reactive, Diatomic). X and Y are:

A. noble gases, oxygen group

B. alkali metals, halogens

C. noble gases, halogens

D. alkali metals, oxygen group

Q.20 The ion of an element, Z^–3 has atomic mass 31 and atomic number 15. How many valence electrons are present in element Z?

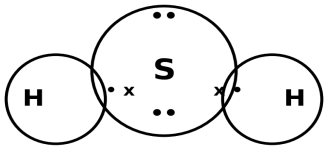
A. 5

B. 8

C. 12

D. 15

Q.21 The dot and cross structure for the molecule of hydrogen sulphide (H2S) is



A. Structure A

B. Structure B

C. Structure C

D. Structure D

Q.22 Which of the following properties is TRUE for ionic compounds?

A. They are malleable and ductile.

B. They have low melting and boiling points.

C. They are brittle and dissolve easily in non-polar solvents.

D. They dissolve in polar solvents and conduct electricity in molten form.

Q.23 Compared to ionic and covalent bonds, intermolecular forces of attractions are

A. weaker than both types of bonding.

B. stronger than both types of bonding.

C. stronger than ionic bonds but weaker than covalent bonds.

D. stronger than covalent bonds but weaker than ionic bonds.

Q.24 Atoms P and Q, belonging to the 2nd period of the modern periodic table, have 1 and 6 electrons in their valence shells respectively. The compound formed is:

A. PQ

B. PQ2

C. P2Q

D. P3Q

Q.25 11_5 B and 27_13 Al are in the same group of the modern periodic table. Hence, they both have the same number of

A. protons.

B. neutrons.

C. inner electrons.

D. valence electrons.

Q.26 On chemical combination, which of these elements can form triple covalent bonds?

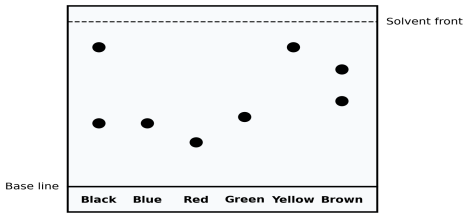
A. Two atoms of oxygen

B. Two atoms of nitrogen

C. Two atoms of sulphur and one atom of carbon

D. Two atoms of oxygen and one atom of carbon

Q.27 Based on the given chromatogram of inks, it can be inferred that:



A. red and blue inks are present in brown ink.

B. yellow and blue inks are present in black ink.

C. black is a mixture of blue, red and green inks.

D. brown is a mixture of red, green and black inks.

Q.28 Which of the following is NOT an example of aqueous solution?

A. Black tea

B. Sea water

C. Orange juice

D. Whipped cream

Q.29 What is the molarity of a solution containing 32 g of potassium sulphate (K2SO4) in 350 cm3 of solution? (Molar mass = 174 g/mol)

A. 0.06 M

B. 0.09 M

C. 0.52 M

D. 64.37 M

Q.30 Sodium chloride crystals can be separated from sand using four processes in the order of:

A. filter -> evaporate -> dissolve -> crystallise

B. dissolve -> filter -> evaporate -> crystallise

C. dissolve -> evaporate -> crystallise -> filter

D. filter -> dissolve -> crystallise -> evaporate

Q.31 If 16 g of sodium chloride is dissolved in water to prepare 200 cm3 of saline solution, the mass/volume (m/v) percentage will be:

A. 0.07%

B. 0.08%

C. 7.41%

D. 8.00%

Q.32 Classification based on solubility in water: Soluble: K2SO4; Insoluble: Mg(OH)2, CCl4. Correct combinations are:

A. I and II.

B. I and III.

C. II and IV.

D. III and IV.

Q.33 A fizzy (carbonated) drink is which type of solution?

A. Solute: Liquid | Solvent: Liquid

B. Solute: Gas | Solvent: Liquid

C. Solute: Gas | Solvent: Gas

D. Solute: Liquid | Solvent: Gas

Q.34 Galvanising is one of the methods to prevent iron from rusting. This method involves coating iron with a protective layer of

A. tin.

B. zinc.

C. aluminium.

D. magnesium.

Q.35 When molten sodium chloride is electrolysed, the chemical equation that shows the change taking place at cathode is

A. Na(s) - 2e- -> Na2+(l)

B. Na2+(l) - 2e- -> Na(s)

C. Na(s) + e- -> Na+(l)

D. Na+(l) + e- -> Na(s)

Q.36 An electrolytic cell does NOT show the use of

A. inert metals as electrodes.

B. a battery to supply energy.

C. salt bridge to control the migration of ions.

D. single electrolyte for the conduction of ions.

Q.37 Metal M placed in solution of metal X ions: solid M decreases while solid X deposits. Correct representation of M and X is:

A. M: iron, X: aluminium

B. M: iron, X: sodium

C. M: zinc, X: copper

D. M: zinc, X: magnesium

Q.38 When an aqueous solution of sodium chloride is electrolysed, the substance that is produced at cathode is

A. oxygen gas.

B. chlorine gas.

C. hydrogen gas.

D. sodium metal.

Q.39 To prevent underground iron pipes from rapid corrosion, blocks of metal X are buried and connected by wires (sacrificial protection). Metal X is:

A. tin.

B. lead.

C. copper.

D. magnesium.

Q.40 Which of the following is an example of electrolyte?

A. Benzene

B. Sulphuric acid

C. Distilled water

D. Carbon tetrachloride

Q.41 [ATP] A student separated a mixture of ammonium chloride and iron filings using a magnet. This is because iron is

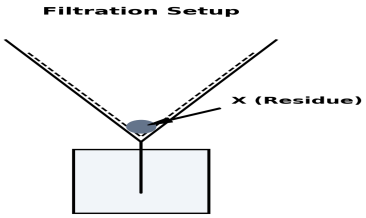
A. highly electronegative.

B. highly electropositive.

C. a metallic solid.

D. an ionic solid.

Q.42 [ATP] A suspension is filtered using a funnel and filter paper. Substance X that remains on the filter paper is identified as the



A. filtrate.

B. residue.

C. crystals.

D. precipitates.

Q.43 [ATP] In a mixture of sand and sugar, the MOST appropriate sequence of steps to obtain pure sample of sugar is:

A. I and III

B. I and IV

C. I, II and III

D. I, II and IV (Dissolution, Filtration, Crystallisation)

Q.44 [ATP] For an experiment, Sarah used a bottle of sodium carbonate solution labelled 'primary standard'. This label indicates that the solution:

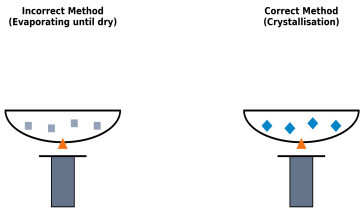
A. is hygroscopic in nature.

B. requires standardisation before its use.

C. is prepared from a secondary standard reagent.

D. has a strength which remains constant with time.

Q.45 [ATP] Evaporating the solution in the china dish until it is dry is an incorrect method because



A. blue crystals of copper sulphate are formed in uneven shapes.

B. blue crystals of copper sulphate decompose into dull white-grey powder.

C. solubility of copper sulphate increases with temperature.

D. solubility of copper sulphate decreases with temperature.

Q.46 [ATP] Once crystals start forming, heating is stopped and the solution cools slowly. This slow crystallisation facilitates in yielding:

A. deep violet crystals.

B. larger, highly pure crystals.

C. smaller, needle like crystals.

D. fluorescent greyish blue crystals.

Q.47 [ATP] Copper sulphate is highly soluble at high temp and decreases appreciably at low temp. Crystallisation can be inferred as a process of:

A. purifying substances.

B. increasing solubility.

C. decreasing solubility.

D. identifying substances.

Q.48 [ATP] Altamash is expected to prepare 0.5 M solution of sodium carbonate in a 100 mL flask. Mass required is (Na=23, C=12, O=16):

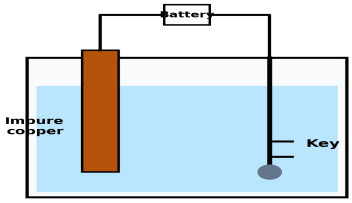
A. 0.53 g.

B. 2.55 g.

C. 5.3 g.

D. 50 g.

Q.49 [ATP] The MAIN purpose of the electroplating method shown (copper onto iron key) is to



A. purify the key.

B. make the key tensile.

C. save the key from corrosion.

D. increase the thickness of the key.

Q.50 [ATP] During electroplating, electrodes made of metals are used because they primarily support the flow of

A. ions.

B. volts.

C. atoms.

D. electrons.